Next week: Midterm (Thurs Feb 12).

Coverage: • Lectures 1–8 (ch. 1 in book + martingales & h-transform).

Today: chapter 2 in book: Poisson processes

First example of continuous-time stochastic process.

- exponential random variables & their properties:
  - memoryless
  - distribution of minimum & minimizer of independent exponentials.
  - distbtn. of sum of i.i.d. exponentials.

- Poisson process:

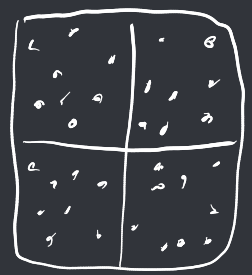
- definition in terms of exponentials
- characterization as point process
  w/ independent Poisson distributed "counts".
- other properties: superposition, thinning?

What is it for?

Poisson point process models "random independent points appearing @ a certain intensity over space".

Think: raindrops on pavement when it starts raining.

1 dim.

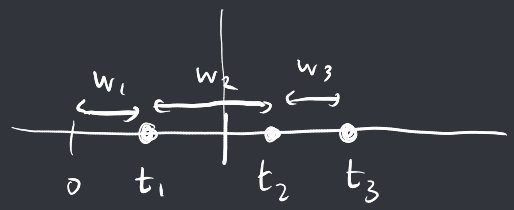


2 dim.

Definition of Poisson point process:

let (w_i) i.i.d. $\underbrace{\text{exponential}(\lambda)}_{P(w_i > t) = e^{-\lambda t}}$.

let $t_n = \sum_{i=1}^{n} w_i$; (t_n) is a P.p.p. on $[0,\infty)$.



Characterization: let $X(a,b) = \#$ points in $[a,b]$.

Then for disjoint $[a_i, b_i]$, $i = 1, \dots, k$,

(i) $\left(X(a_i, b_i)\right)_{i=1}^{k}$ are independent

(ii) & $X(a_i, b_i) \overset{d}{=} \text{Poisson}\left(\lambda(b_i - a_i)\right)$.

if (i) holds, (ii) is implied by

$$E\left[X(a,b)\right] = \lambda(b-a) \quad \forall \quad a<b.$$

- Independence property for particle counts corresponding to (tn) will follow from memoryless property on exponential.

- Poisson distribution is deduced by subdividing into small intervals and using

$$\text{Binomial}\left(n, \lambda(b-a)/n\right) \xrightarrow{n\to\infty} \text{Poisson}(\lambda(b-a)).$$

Exponential random variables

$$w \overset{d}{=} \exp(\lambda) \quad \text{if} \quad P(w>t) = e^{-\lambda t}; \quad t \geq 0.$$

mean & variance: $\exp(\lambda)$ has density function

$$f(t) = \frac{d}{dt} P(w \le t) = \frac{d}{dt}\left(1 - e^{-\lambda t}\right)$$

$$= \lambda e^{-\lambda t}$$

so

$$E[w] = \int_0^\infty t \lambda e^{-\lambda t} \, dt$$

$$= \dots = 1/\lambda$$

variance has $\mathrm{Var}(w) = 1/\lambda^2$

(compute $E[w^2] - (E[w])^2$)

memoryless property

$$P(w > t+s \mid w > t) = P(w > s) \qquad \forall s,t \geq 0.$$

w : time to wait until particle decays

exponential distbtn. has "hazard rate"

$$h(t) := -\frac{d}{dt} \log P(w > t) = -\lim_{s \to 0} s^{-1} P(w > t+s \mid w > t)$$

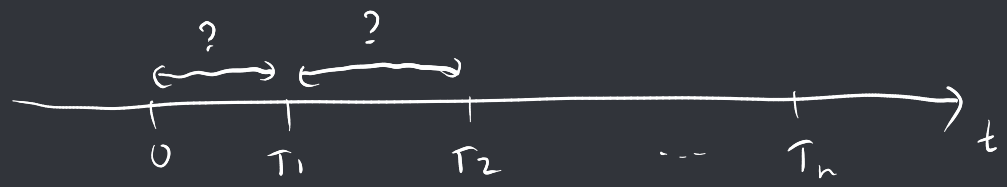
constant w.r.t. t.

distribution of minimum & minimizer

<u>ex</u>: n particles in box. Each one decays independently @ rate λ.

Find $E[\underbrace{\text{time until none remain}}]$.

$= T_n$ where T_k = time k^{th} particle decays



On one hand $T_n = \max(W_1, \dots, W_n)$

where W_i is decay time of ith particle.

Doesn't help to compute $E[T_n]$.

Instead let $U_k = T_k - T_{k-1}$.

<u>Claim</u>: $(U_k)_{k=1}^{n}$ independent, $U_k \overset{d}{=} \exp(\lambda(n-(k-1)))$

<u>Required property</u>: if $V_1, \dots, V_n$ indep. & $V_i \overset{d}{=} \exp(\lambda_i)$

then $\min(V_1, \dots, V_n) \overset{d}{=} \exp(\lambda_1 + \dots + \lambda_n)$

case $n=2$:

$$P(\min(V_1, V_2) > t)$$

$$= P(V_1 > t, V_2 > t)$$

$$= P(V_1 > t)\, P(V_2 > t)$$

$$= e^{-\lambda_1 t}\, e^{-\lambda_2 t}$$

$$= e^{-(\lambda_1 + \lambda_2) t}.$$

sim. for larger n.

Using this property, if $n-(k-1)$ particles remain then time of next decay = min of $n-(k-1)$ exponentials each w/ rate λ

$$\overset{d}{=} \exp(\lambda(n-(k-1)))$$

(uses memoryless property)

so $$E[U_k] = \frac{1}{\lambda(n-(k-1))}$$

& $T_n = U_1 + \cdots + U_n$ so $E[T_n] = \frac{1}{\lambda}\left(\frac{1}{n} + \frac{1}{n-1} + \cdots + 1\right)$

$\sim \frac{1}{\lambda} \log n.$

distribution of minimizer.

Let $W_1, \dots, W_n$ indep. $W_i \overset{d}{=} \exp(\lambda_i)$

then $$P(W_i = \min_k W_k) = \frac{\lambda_i}{\lambda_1 + \dots + \lambda_n}. \quad (*)$$

ex: if $W_1 = \exp(1)$, $W_2 = \exp(2)$ $\quad P(W_1 < W_2) = 1/3$

(*) Proof in book.

Moreover, if $I = \operatorname{argmin}\{W_i : i = 1, \dots, n\}$

then I & W_I are independent.

(in book)

<u>ex</u>: if W_1, W_2 as above then

$$P(W_1 < W_2 \ \& \ W_1 > t) = P(I = 1 \ \& \ W_I > t)$$

$$= \tfrac{1}{3} \cdot e^{-3t}$$

Sum of exponentials

if $W_i \overset{d}{=} \exp(\lambda)$, indep, then

$W_1 + \cdots + W_n$ has density function

$$\lambda e^{-\lambda t} \frac{(\lambda t)^{n-1}}{(n-1)!}$$

(do by induction in n)

$n=2$ case:

$$P(W_1+W_2>t) = \int_0^\infty \lambda e^{-\lambda s} \underbrace{P(W_2>t-s)}_{\substack{=1 \text{ if } s>t \\ \text{otherwise } = e^{-\lambda(t-s)}}} \, ds$$

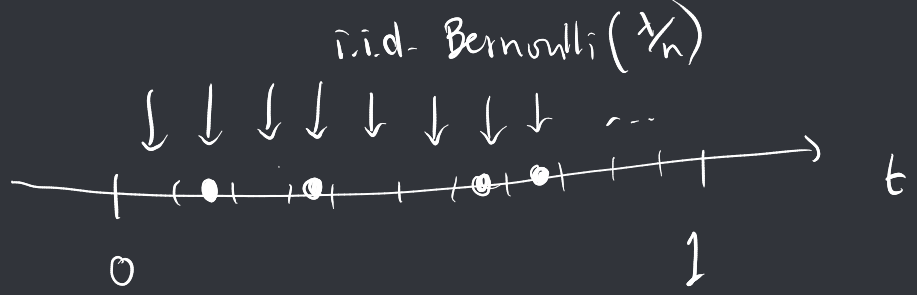
$\left(\vdots\right)$

Poisson point process,
construction by approximation & subdivision.

Definition given earlier as partial sum of

i.i.d. sequence of exponentials.

Can construct approximately using i.i.d. Bernoulli.



- subdivide into n subintervals of length $1/n$.
- to have $E[\#\text{ in } (a,b)] \approx \lambda(b-a)$ put a point in k^{th} subinterval w.p. $p = \lambda/n$

this has $\approx$ independent particle counts over disjoint intervals

and E # particles is $\approx$ correct.

as $n \to \infty$ does this converge in dist$^{\underline{bn}}$ to Poisson point process?

Total # particles in $[0,1] \overset{d}{=} \text{Binomial}(n, \lambda/n)$

as $n \to \infty$ $\underbrace{\text{Binomial}(n, \lambda/n)}_{N_n} \longrightarrow \text{Poisson}(\lambda)$.

pf:

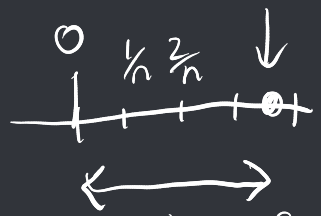
$$P(N_n = k) = \underbrace{\binom{n}{k}\left(\lambda/n\right)^k}_{= \frac{n!}{k!(n-k)!} \frac{\lambda^k}{n^k}} \underbrace{\left(1 - \lambda/n\right)^{n-k}}_{\to e^{-\lambda} \text{ as } n \to \infty \text{ for fixed } k}$$

$$\underbrace{\frac{n(n-1)\cdots(n-k+1)}{n^k}}_{\to 1 \text{ as } n \to \infty} \frac{\lambda^k}{k!}$$

Distribution of $X(0,1)$ converges to Poisson(λ)

Similar argument shows $X(a,b) \longrightarrow \text{Poisson}(\lambda(b-a))$

What about distance between points?



how far till first point? $\overset{d}{=} \frac{1}{n}\overbrace{\text{Geometric}\left(\frac{\lambda}{n}\right)}^{S}$

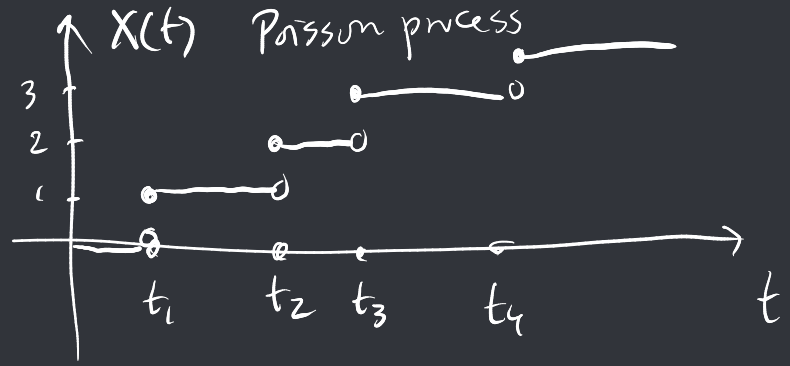
$$P\left(\tfrac{1}{n}S > t\right) = P\left(S > \lfloor nt \rfloor\right) = \left(1-\tfrac{\lambda}{n}\right)^{\lfloor nt \rfloor}$$

$$\xrightarrow{n\to\infty} e^{-\lambda t}$$

Poisson point process is a random set of points.

Poisson (counting) process takes the cumulative sum of particle counts:

if (t_n) is PP.P.



in earlier notation $X(t) = X(0,t)$

$= \#$ particles in $[0,t]$